BATTLEBEAM

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Introduction

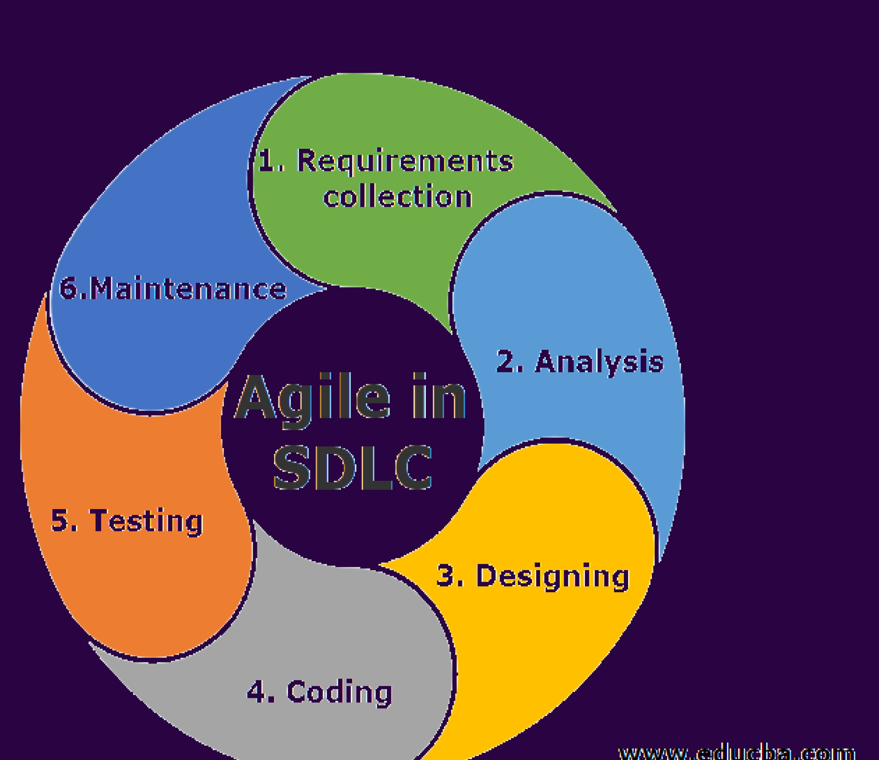
Laser tag is a popular interactive game that combines physical activity, teamwork, and competition. However, traditional laser tag systems are usually limited to commercial venues, making them expensive and inconvenient for regular use. Existing home alternatives often lack real-time interaction, reliability, or advanced game control. To address these limitations, BattleBeam was designed as a portable, home-based laser tag system that utilizes infrared technology, IoT communication, and a mobile application. The system enables players to enjoy a complete laser tag experience indoors with real-time scoring, team management, and synchronized gameplay through affordable and accessible hardware.

Objectives

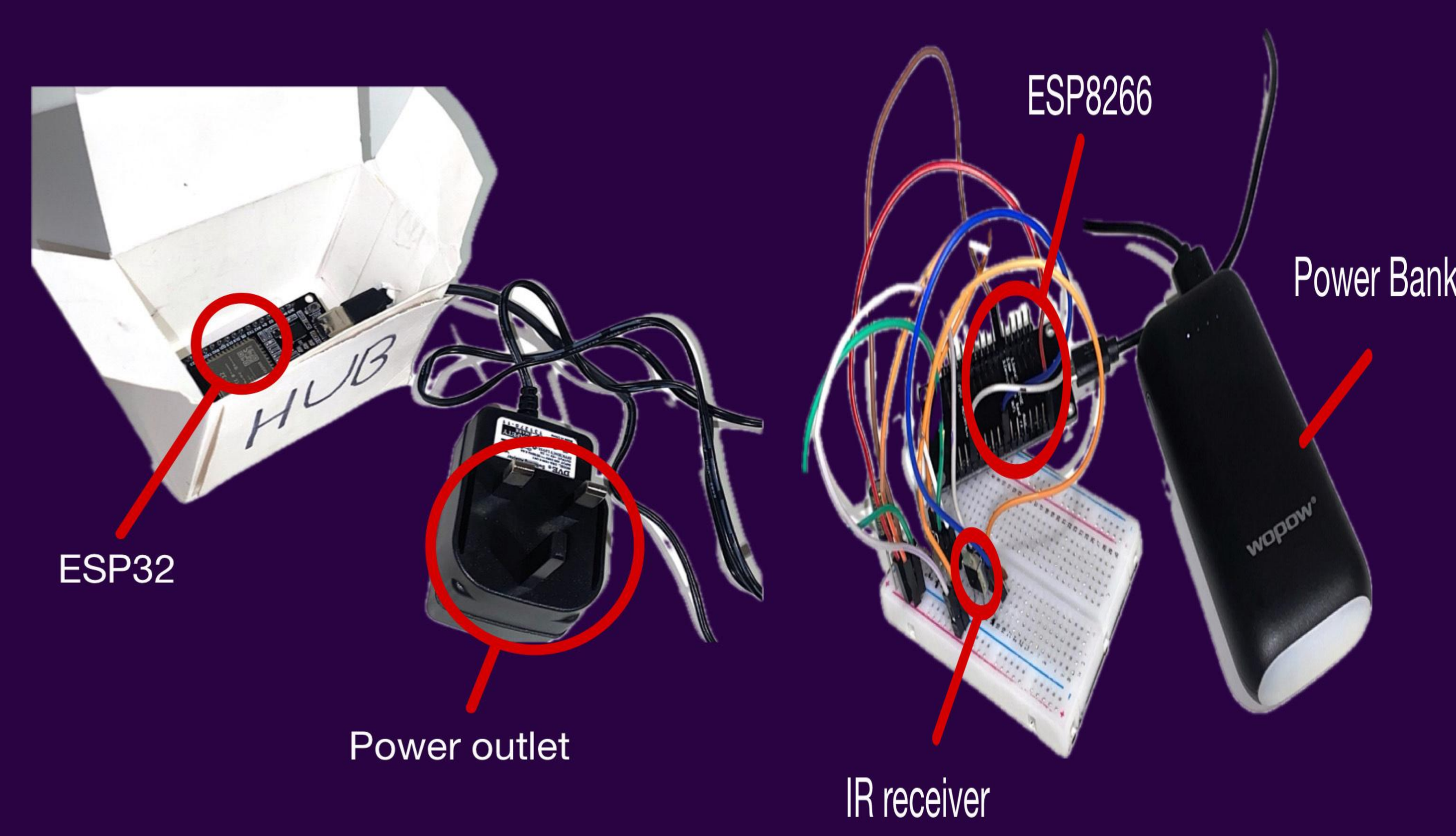
- Develop IR-based laser guns using ESP32
- Develop wearable vests using ESP8266 for hit detection
- Implement a central ESP32 hub for game control and scoring
- Create a cross-platform mobile app for game management
- Enable real-time score tracking and team-based gameplay
- Ensure the system is portable, rechargeable, and easy to use

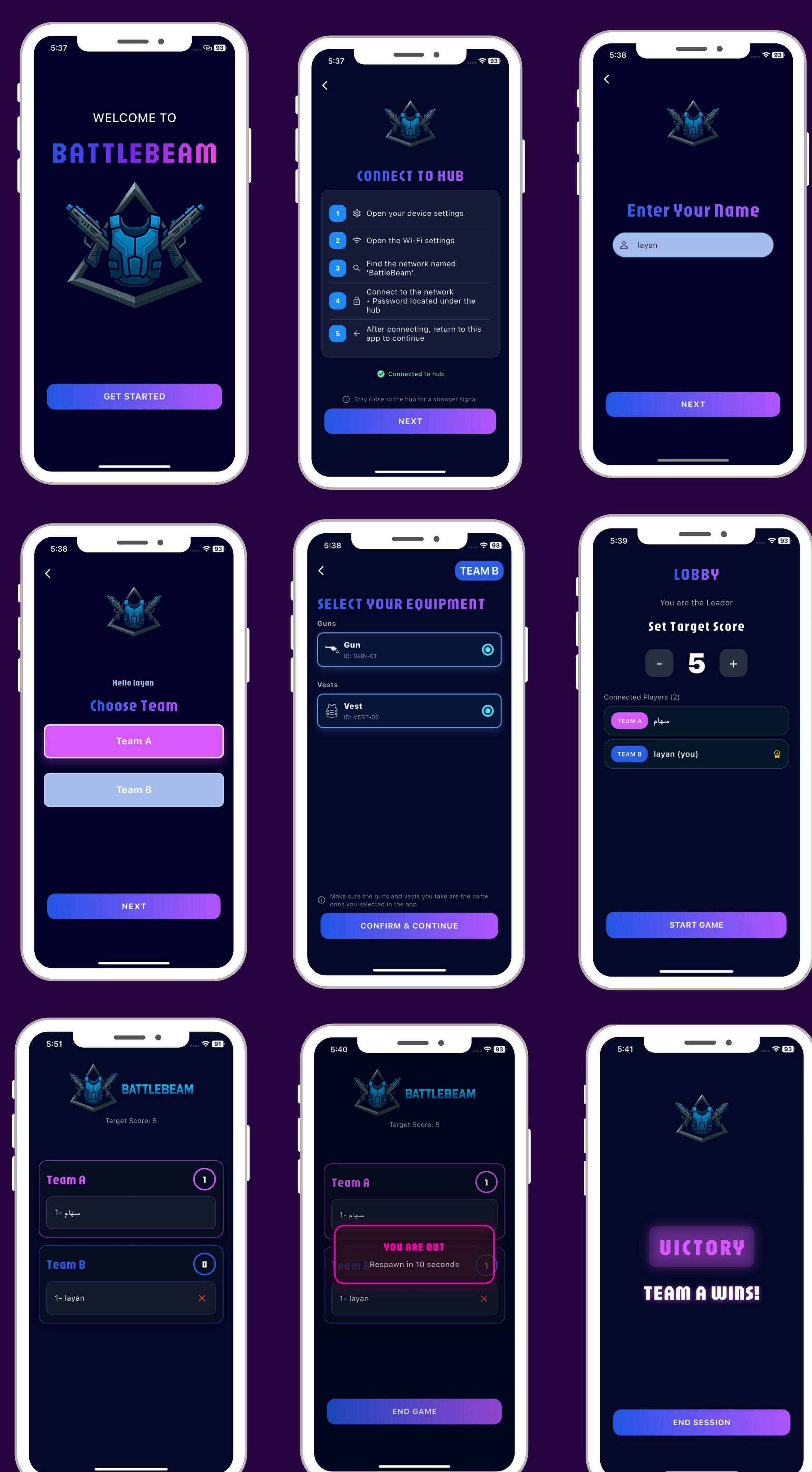
Methodology

The Agile methodology was employed to guide the iterative development and testing for BattleBeam home-based laser tag system.



Results





Conclusion and Future work

Conclusion:

- BattleBeam successfully delivers a functional home-based laser tag experience
- Achieved real-time communication, accurate hit detection, and live scoring
- Central hub ensured fair gameplay and synchronization
- Testing confirmed reliability, usability, and system stability
- The project met all defined functional and non-functional requirements

Future Work:

- Add more game modes to increase engagement.
- Add RGB feedback
- Support more players and extended play areas
- Introduce different weapon types (grenades, bazooka)

References

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5. L. Atzori et al., "The Internet of Things: A Survey," Computer Networks, 2010.

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